

Causal forests and more

Lecture 10 – AI4H

Andreas Bjerre-Nielsen

Where are we at?

- Part I: Foundations
 - Tree-based, deep learning, transformers, LLMs
- Part II: Policy and ML
 - *Fairness*
 - Prediction policy
 - AI for policy, other uses
- Part III: Causal ML
 - Causal trees and forests
 - Double ML

Takeaways last time

- Evidence of positive impact of AI on decisions
 - Admission screening
 - Workplace productivity (customer support)
- However..
 - Not for medical diagnosis when used by humans
 - Problems
 - Automation neglect
 - Signal dependence neglect

Overview of this lecture

- Delegation in prediction policy
- Causal inference refresher
- Garden-based causal inference
 - Causal Tree
 - Causal Forest
 - Generalized Random Forest

Delegation in prediction policy

A refresher

Delegation framework

- Who decides prediction?
- We make rule that decides whether to collect human assessment
 - If AI is sufficiently confident – use its predictions
 - Otherwise
 - > Collect human assesment and use that (without AI)
 - > Or collect human assesment with AI if expected benefits outweigh increased processing costs
- Why does this matter
 - Accuracy: who is more likely to be right?
 - Cost/time: who can decide faster or cheaper?
 - Cognitive biases: how does human behavior change when assisted by AI?

Agarwal et al. (2024)

- Optimal allocation rule:
 - AI diagnoses cases with high-confidence predictions.
 - Humans diagnose cases with valuable contextual information (e.g., patient history).
 - Human + AI mode is rarely optimal under existing belief-updating biases.

Agarwal, Moehring & Wolitzky (2025)

- Goal: Derive a general rule for when to automate, delegate, or assist without modeling full human psychology.
- Binary classification tasks:
 - AI predicts: $P(\text{correct label} \mid x)$
 - Human observes AI's prediction and may act or defer
- Designer chooses:
 - how much AI info to reveal
 - which cases to automate
- Objective: maximize expected accuracy

Optimal delegation

- The Sufficient Statistic $V(x)$
 - $V(x)$: probability a human classifies correctly when seeing AI prediction x
 - Captures *all* behavioral & effort effects
 - → Once $V(x)$ is estimated, it's sufficient to compute the optimal policy
 - No need for full structural model of human decision-making

Experimental Validation

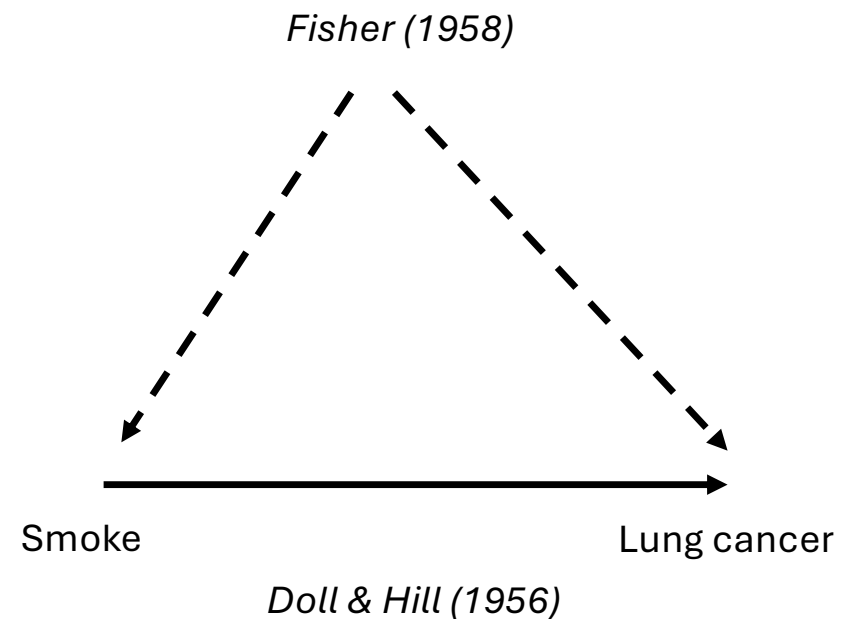
- Two-stage fact-checking experiment
 - Stage 1: estimate $V(x)$
 - Stage 2: test predicted optimal policies
- Findings:
 - No-Disclosure + Automation is best $\rightarrow$ little added value from joint use
 - When automation not allowed $\rightarrow$ Full Disclosure + No Automation best
- Predictions within 1.6 pp of actual accuracy $\rightarrow$ model validated
- Intuition / behavioral Insights
 - Humans under-respond to AI (confirming radiology findings)
 - Cause: overconfidence in own signal, not distrust of AI
 - AI information crowds out effort, but effect is minor
- Collaboration value $\approx$ zero $\rightarrow$ key is *smart delegation*, not “AI teaming”

Causal inference

A refresher

What is causality?

- Relationship between two or more variables such that whereby a change in one or more variable(s) *affect(s)* the distribution of one or more other variable(s).
- We can draw these relationships (from *The Book of Why*, Judea Pearl), e.g. smoking example.
- Ronald Fisher argued that unobserved confounders could cause smoking and lung cancer



Establishing causality

- Currently there are two broad approaches for establishing causal relationships:
- Experiment and quasi-experiments
 - Leverage randomness
- Structural equation models
 - Used for structural econometric choice models etc.
 - Also used estimating causal graphs, e.g. as by Judea Pearl

Potential outcomes

Rubin Causal Model

- Denote the treatment variable as D_i where $D_i = 1$ corresponds to unit i being treated, while $D_i = 0$ is not treated. Define the potential outcomes:

$$Y_i = \begin{cases} Y_i(1), & D_i = 1 \\ Y_i(0), & D_i = 0 \end{cases}$$

- The observed outcome Y_i can be written in terms of potential outcomes as

$$Y_i = Y_i(0) + \underbrace{[Y_i(1) - Y_i(0)]}_{\text{(average) treatment effect of } D_i \text{ on } Y_i} \cdot D_i$$

- But we never observe the same individual i in both states. This is the **fundamental problem of causal inference**.

Selection problem

- We need some way of estimating the state we do not observe (the *counterfactual*)
- Usually, our sample contains individuals from both states - treated and untreated.
- So why not do a naive comparison of averages by treatment status? i.e. $E[Y_i|D_i = 1] - E[Y_i|D_i = 0]$
- We can rewrite into:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0] = E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 1] + \underbrace{E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{\text{selection problem}}$$

- The decomposition:
 - Average treatment effect (ATE)
 - Selection: difference in average $Y_i(0)$ between the two groups. Likely to be different from 0 when individuals are allowed to self-select into treatment. Often referred to as *selection bias*.

Random assignment solves the problem

Random assignment implies is independent of potential outcomes

Selection bias term is zero: $E[Y_i(0)|D_i = 1] = E[Y_i(0)|D_i = 0]$

- Intuition: non-treated individuals can be used as counterfactuals for treated (*what would have happened to individual i had he not received the treatment*)
- Solves the fundamental problem of causal inference
- Can be done by research, e.g., Randomized Control Trials or survey experiments

An alternative to experiments

- Quasi-experiments: randomization happens as "by-product"
 - Matching (*recap today*)
 - Differences in Differences
 - Regression Discontinuity Design
 - Instrument variables

Matching estimators

The what and why of matching

- **What** - we construct counterfactual potential treated and control units.
 - We *match* observations across treatment and control based on similarity.
- **Why** - matching control for used covariates
 - excludes (observable) confounders
 - may improve precision of treatment estimate of experiments (less variance)
- Note: An alternative to matching is to using regression - basically same idea.
- Problem: unobserved factors or endogeneity may still confound

How to match

- We use a set of covariates X for matching.
- Two core ideas:
 - We match on (observable) covariates
 - We require sufficient similarity by some metric over covarities
 - We match on propensity
 - We require sufficient similar probability of treatment (prediction)

Covariate-based matching

- Exact matching
 - We match a treatment i obs. with control obs. j if
 - $X_i = X_j$, i.e. they are exactly identical covariates,
 - same as $\|X_i - X_j\|_2 = 0$, i.e. zero Euclidian distance
 - We can estimate the treatment effect locally for each covariate combination and aggregate
- Problem requires all observations are matched to get balance
 - Can be handled by near-exact matching, e.g.,
 - Coarsened Exact, Radius/K-nearest

Propensity score matching

- Alternative way of match on likelihood of treatment.
- Procedure:
 1. estimate a model that predicts treatment
 2. match with observations of similar treatment likelihood
(use match function, e.g. nearest neighbor, caliper/radius)
 3. compute counterfactual outcomes for treatment and control
 4. (possibly adjust for differences in observed covariates)
 5. compute ATE

Summary matching

- Useful tool, but requires that we know all relevant factors
 - can be useful to minimize variance of experimental estimates
 - problem in observational studies - often there are unobserved confounders and selection
- If we think there is unobserved selection or endogeneity:
 - Use quasi-experimental methods which can handle this, e.g. diff-in-diff or regression discontinuity

Garden-based causal inference

Introduction and overview

Average Joe and beyond

- Suppose, we have credible measures of average treatment effect, τ
 - Can we get personalized estimates?
 - Measure whether certain groups are affected differently by our new school policy
 - e.g. boys vs. girls, natives vs. immigrants
- Some react positively to one kind of information, others to another
- Ideas?

Beyond average Joe

- Conditional Average Treatment Effects (CATE)
- Treatment effect for given characteristics x
$$\tau(x) = E[Y_i(1) - Y_i(0)|X = x]$$
- Methods exist, e.g. use regression analysis - *interaction effects*.
- But.. **True model is unknown..!**
 - May need to test model on data.
 - Can lead to conclusions based on data mining (dangerous!!)

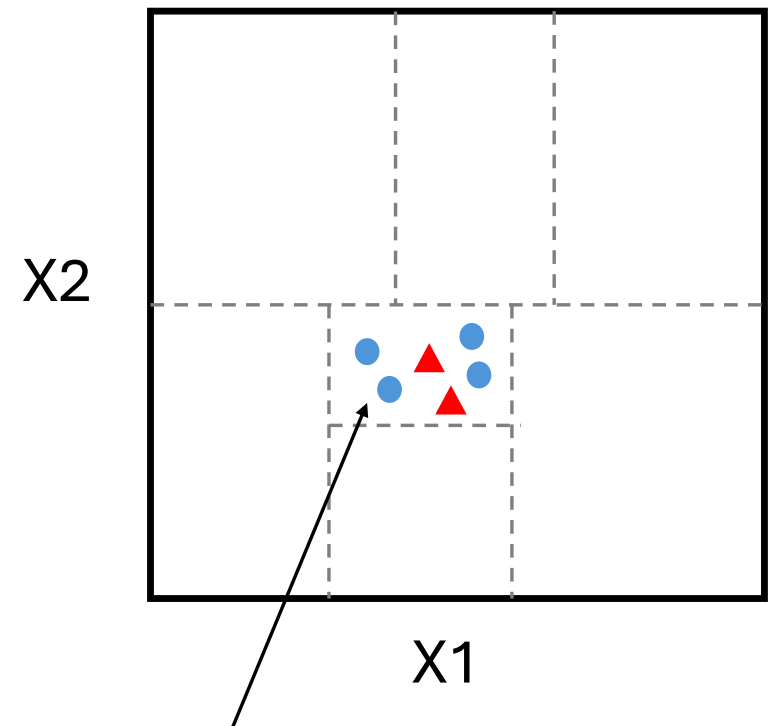
A magic tree

Adaptive approach

- A data driven approach
 - use all data for training decision tree
 - Partition all of covariate space X into categories based **outcome** similarity
 - enough treatment and control in each leaf
 - then estimating treatment effects in covariate partitions
 - measure treatment effects in each partition group (=leaf in tree model)

→ heterogeneous treatment effect, $\tau(x)$

Illustration of partitioning



Partition/category

→ estimate treatment effect here as outcome difference between red/blue

Quiz on adaptive approach

- **Quiz:** is this different from propensity scores?
 - Propensity scores has treatment assignment D_i as target.
 - The adaptive approach uses outcome y_i as target.
- **Bonus quiz:** Are estimates credible?

Honest approach

- Could we use out-of-sample intuition?
- [Athey and Imbens \(2016\)](#) suggest to let data speak **honestly**:
 - half of sample ($\mathcal{S}^{\text{tr}}$) for training decision tree
 - partitions X into categories based outcome similarity
 - enough treatment and control in each leaf
 - other half ($\mathcal{S}^{\text{est}}$) for estimating treatment effects
 - measure treatment effects in each partition group (=leaf in tree model)
- This is similar to splitting into train and test
 - prevents data-leakage
 - allows honest evaluation of model performance!

Core assumption

- Potential outcomes and treatment assignment are unconfounded given covariates

$$D_i \perp\!\!\!\perp (Y_i(1), Y_i(0)) \mid X$$

- where $\perp\!\!\!\perp$ is a symbol for conditional independence (strong assumption!!)
 - recall from earlier
 - always holds for experiments
 - or well-specified propensity scores (note: assumption cannot be tested)

Standard vs. causal tree

- The usual way of training decision trees is Classification And Regression Trees (CART).
 - Splits leaves repeatedly based on criteria (e.g. entropy, MSE)
 - We can put in restriction, e.g. depth of trees (hyperparameters)
- Causal trees
 - new criteria:
 - expected MSE (in hypothetical test set): $E[\underbrace{(Y_i - \hat{Y}_i)^2}_{= \text{MSE}} - Y_i^2]$
 - idea: new term Y_i^2 penalizes small leaves
 - note: same ranking as MSE, matters for properties

Tree performance

Simulation exercise of statistiscal performance

Table 1. Simulation study

$N^{tr} = N^{est}$ Estimator	Design 1		Design 2		Design 3	
	500	1,000	500	1,000	500	1,000
	No. of leaves					
TOT	2.9	3.2	2.9	3.5	3.6	5.4
F-A	6.1	13.1	6.3	13.0	6.2	13.0
TS-A	4.0	5.4	3.4	5.1	3.4	6.6
CT-A	4.0	5.5	3.2	3.7	3.5	5.4
F-H	6.0	12.9	6.3	13.0	6.3	13.1
TS-H	4.3	7.8	5.6	11.4	5.9	12.4
CT-H	4.2	7.6	5.6	11.4	6.1	12.5
	Infeasible MSE divided by infeasible MSE for CT-H*					
TOT-H	1.554	1.938	1.089	1.069	1.081	1.042
F-H	1.790	1.427	1.983	2.709	1.502	2.085
TS-H	0.971	0.963	1.183	1.145	1.178	1.338
	Ratio of infeasible MSE: Adaptive to honest†					
TOT-A/TOT-H		1.021		0.754		0.717
F-A/F-H		0.491		0.985		0.993
T-A/T-H		0.935		0.841		0.918
CT-A/CT-H		0.929		0.851		0.785
	Coverage of 90% confidence intervals – adaptive					
TOT-A	0.82	0.85	0.78	0.81	0.69	0.74
F-A	0.89	0.89	0.83	0.84	0.82	0.82
TS-A	0.84	0.84	0.78	0.82	0.75	0.75
CT-A	0.83	0.84	0.78	0.82	0.76	0.79
	Coverage of 90% confidence intervals – honest					
TOT-H	0.90	0.90	0.90	0.89	0.89	0.90
F-H	0.90	0.90	0.90	0.90	0.90	0.90
TS-H	0.90	0.90	0.91	0.91	0.89	0.90
CT-H	0.89	0.90	0.90	0.90	0.89	0.90

* $MSE_{\tau}(S^{te}, S^{est}, \pi_{Estimator}(S^{tr})) / MSE_{\tau}(S^{te}, S^{est}, \pi_{CT-H}(S^{tr}))$.

† $MSE_{\tau}(S^{te}, S^{est} \cup S^{tr}, \pi_{Estimator-A}(S^{est} \cup S^{tr})) / MSE_{\tau}(S^{te}, S^{est}, \pi_{Estimator-H}(S^{tr}))$.

Usefulness of causal trees

- Two approaches to measuring heterogeneity
 - Data driven heterogeneity (causal trees)
 - A priori sensible heterogeneity
 - e.g. gender, socioeconomic, ethnicity
 - we use regression model and have interaction with desired variable
- When to choose which?
 - Choose data driven heterogeneity for policy where you want to maximize impact given data (no theory)
 - If we want to test whether certain subgroups are adversely affected in more than one dimension

Summary of causal tree

- Leverage machine learning idea:
 - Heterogeneity is estimated separate from treatment effects.
 - New scoring function makes smaller leafs.
 - Outperforms adaptive procedure
- Main advantage
 - Structure of heterogeneity from data.
 - Can be part of pre-analysis plan - only one solution (given split of data!).

Limits to causal trees

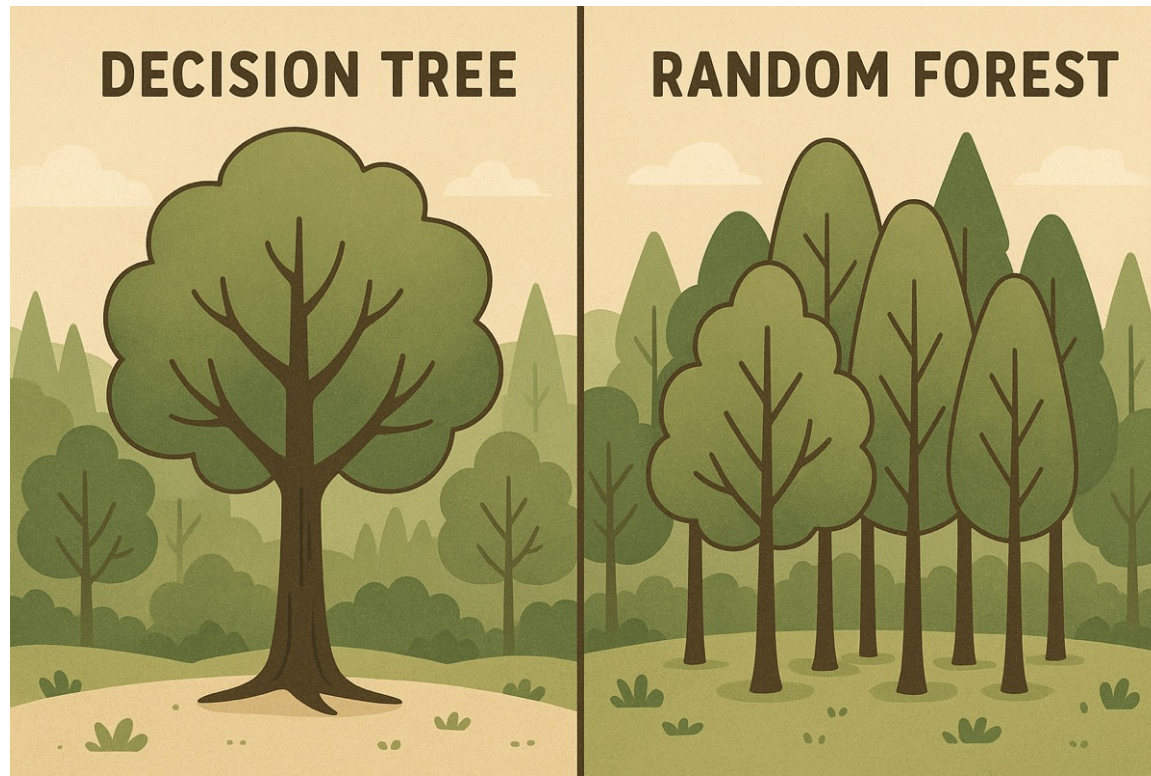
- Only usefulness with random assignment
 - No handling of unconfounding!
- No asymptotic properties
 - No large sample convergence of estimators
 - → *Inconsistent estimates!!*

Strength in numbers

From causal trees to forests

(decision) Tree: iteratively splits data into subsets (partitions) and calculates mean outcome in leaves (end of splits)

> Minimize on some criteria, often entropy or similar loss function



Highly flexible model,
but high **overfitting**

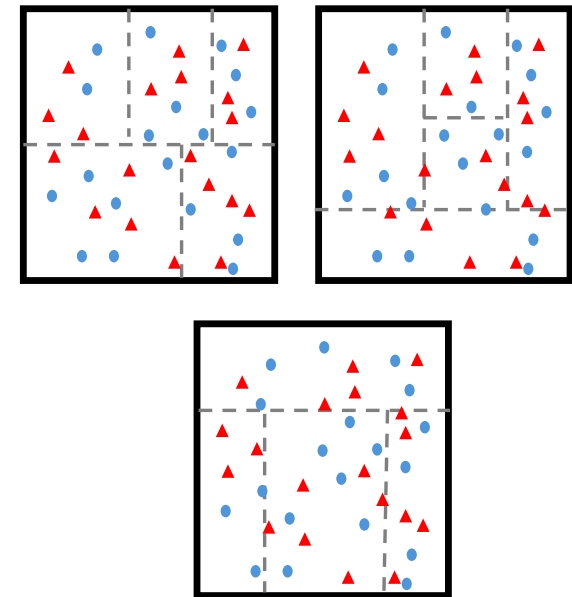
Curbs overfitting by making
many different trees and aggregating

(random) Forest:
Collection/ensemble of
decision trees

> Subset of data by
bootstrap (sampling with
replacement)
> Subset of features

Random forest for inference and treatment effects

- **Causal Trees:** The goal of causal trees is to establish unbiased, consistent estimates of heterogeneous treatment effects
 - also known as conditional average treatment effects (**CATE**)
 - the effect size is denoted $\hat{\tau}(x)$;
 - standard tools for inference, e.g. using statistical tests locally
- **Causal Forest:** What is the output from the decisions trees? Each tree produces a partitioning of the feature space X .
 - Developed by [Wager and Athey \(2017\)](#)



Double Sample Trees - causal

- For Causal Trees
 - first half ($|\mathcal{J}|, |\mathcal{J}| = \lceil s/2 \rceil$)
 - purpose: train Decision Tree for partitioning
 - objective: minimize adjusted MSE
 - require at least k observations of *both* treatment and control in all leaves of $\mathcal{J}$ -sample
 - second half ($\mathcal{J}, |\mathcal{J}| = \lfloor s/2 \rfloor$)
 - purpose: estimating treatment effects, $\hat{\tau}(x)$

Double Sample Trees - regression

- For Regression Trees
 - first half ($|J|, |J| = \lceil s/2 \rceil$)
 - training Decision Tree
 - objective: minimize MSE / gini / accuracy
 - require at least k observations of in all leaves of J -sample
 - second half ($J, |J| = \lfloor s/2 \rfloor$)
 - estimate outcomes
- **Quiz:** How is this different from normal Decision Trees??
 - > honest estimation – built-in “test/train split”

Econometric properties (1)

- [Wager and Athey \(2017\)](#) show

- We can estimate the variance of CATE

$$\widehat{V}_{IJ}(x) = \frac{n-1}{n} \left(\frac{n}{n-s} \right)^2 \sum_{i=1}^n \text{cov}[\widehat{\tau}_b^*(x), N_{ib}^*]^2$$

- a

Econometric properties (2)

- From Theorem 4.1 in [Wager and Athey \(2017\)](#)
 - The conditional average treatment estimates are unbiased and consistent
 - unbiased: no systematic error of measurement
 - consistency: with more data our estimate approaches true value
- Moreover, we can do inference:
 - The variance estimator $\widehat{V}_{IJ}(x)$ is consistent.
 - Treatment effect estimates are asymptotic normal and unbiased
$$(\hat{\tau}(x) - \tau(x)) / \sqrt{\text{var}[\hat{\tau}(x)]} \Rightarrow \mathcal{N}(0,1)$$
- Caveat: only works for evaluating treatment effects in one point x !
 - Do not perform multiple tests.
 - > Handle with omnibus test

Other useful forests

- Two more procedures using
 1. Double Sampled Trees
 - Regression trees for predicting outcome ($=\hat{\mu}(x)$)
 2. Propensity Trees
 - Using propensity trees for propensity score matching
- What is the shared procedure?
 - Each tree is estimated using repeated subsampling (**no replacement**)
 - Contrast to bootstrap aggregation in random forest (sample **with replacement**)
 - Random subsample of features

More results

- [Wager and Athey \(2017\)](#) show that the same properties of Double Sample Trees using causal trees also hold analogously for regression trees.
- Random forests have the property of being asymptotic normal and can thus be used for inference
 - Similar intuition as idea of nested cross validation where we could do inference

Simulation experiment

- [Athey, Tibshirani, Wager \(2019\)](#) compare causal forest to nearest neighbor methods
 - random forest is kind of local nearest neighbor estimate
 - based on work by Lin and Jeon (2006).

Simulation (1)

- **Simulation setup**

- simulation setup: no treatment effect, only confounding factors
- method: propensity trees
- comparison of estimated treatment effects
 - lower MSE and better coverage
 - coverage falls for increasing number of variables d

d	Mean-squared error			Coverage		
	CF	10-NN	100-NN	CF	10-NN	100-NN
2	0.02 (0)	0.21 (0)	0.09 (0)	0.95 (0)	0.93 (0)	0.62 (1)
5	0.02 (0)	0.24 (0)	0.12 (0)	0.94 (1)	0.92 (0)	0.52 (1)
10	0.02 (0)	0.28 (0)	0.12 (0)	0.94 (1)	0.91 (0)	0.51 (1)
15	0.02 (0)	0.31 (0)	0.13 (0)	0.91 (1)	0.90 (0)	0.48 (1)
20	0.02 (0)	0.32 (0)	0.13 (0)	0.88 (1)	0.89 (0)	0.49 (1)
30	0.02 (0)	0.33 (0)	0.13 (0)	0.85 (1)	0.89 (0)	0.48 (1)

Simulation (2)

- Setup: heterogeneous treatment effect, **no** confounding factors
 - comparison of estimated treatment effects
 - lower MSE and better coverage
 - coverage falls for increasing number of variables d

-

d	Mean-squared error			Coverage		
	CF	7-NN	50-NN	CF	7-NN	50-NN
2	0.04 (0)	0.29 (0)	0.04 (0)	0.97 (0)	0.93 (0)	0.94 (0)
3	0.03 (0)	0.29 (0)	0.05 (0)	0.96 (0)	0.93 (0)	0.92 (0)
4	0.03 (0)	0.30 (0)	0.08 (0)	0.94 (0)	0.93 (0)	0.86 (1)
5	0.03 (0)	0.31 (0)	0.11 (0)	0.93 (1)	0.92 (0)	0.77 (1)
6	0.02 (0)	0.34 (0)	0.15 (0)	0.93 (1)	0.91 (0)	0.68 (1)
8	0.03 (0)	0.38 (0)	0.21 (0)	0.90 (1)	0.90 (0)	0.57 (1)

Summary of causal forest ([Wager and Athey, 2017](#))

- Builds on Causal Trees method
 - Aggregates over many trees
- New properties
 - strong econometric properties
 - unbiased and consistent
 - asymptotic normality given x
 - causal and regression forest allows inference!
- Problems:
 - Choose only one
 - unconfounding (propensity) or
 - estimate CATE
 - Coverage was not good, especially for higher d !

Generalized Random Forest

Leveraging both unconfounding and heterogeneity

A higher aim

- Causal forests are pretty cool. Can we use our honest procedure more generally?
- [Athey, Tibshirani, Wager \(2019\)](#) argue that we can estimate any quantity $\theta(x)$ identified via local moment conditions, e.g.
 - simultaneously unconfound and find heterogeneity?
 - find heterogeneous treatment effects from IV estimation?
- What is a moment condition?
 - Similar to solution to first order condition
 - More general - can incorporate extra restrictions (e.g. unconfounding)

A different purpose

- How does this look?

- The general moment conditions

$$E[\psi_{\theta(x), \nu(x)}(O_i) | X_i = x] = 0, \quad \forall x.$$

- Where ψ is an estimating function, maps parameters and data into moment equations
- Parameters
 - θ parameter we want estimate
 - ν is nuisance we want to "partial out"
- Data
 - O_i main objects we are interested in modelling, e.g. Y_i, D_i using X_i (covariates)

Moment representation

- Suppose we want to estimate conditional average treatment effects
- Functional form:
 - $\psi_{\beta(x), c(x)} = Y_i - \beta(x)W_i - c(x)(1 - W_i),$
- where,
 - β is treatment effect
 - c is nuisance parameter
- Kernel methods can be used to unconfound and compute heterogeneous effects *simultaneously*
 - Problem how to decide weights?

Weighting of three trees

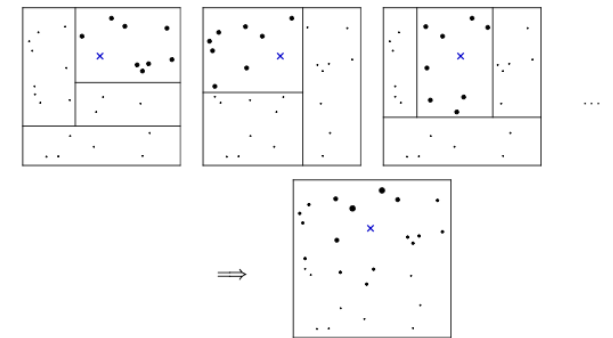


Fig 1: Illustration of the random forest weighting function. The rectangles depicted above correspond to terminal nodes in the dendrogram representation of Figure 4. Each tree starts by giving equal (positive) weight to the training examples in the same leaf as our test point x of interest, and zero weight to all the other training examples. Then, the forest averages all these tree-based weightings, and effectively measures how often each training example falls into the same leaf as x .

(From Athey, Wager, Tibshirani, 2019)

Generalized Random Forest

- [Athey, Tibshirani, Wager \(2019\)](#) show that kernel weights can be estimated using forest methods
 - can be adapted for different purposes
 - quantile regression
 - heterogeneous treatment effects
 - instrumental variables
- How: Use estimating equation, ψ to estimate tree splits iteratively on subsample.
 - A) View forests as a weights of similar neighbors
 - B) Amount of partitions where observations
$$\alpha_i(x) = \frac{1}{B} \sum_{b=1} \frac{\mathbb{1}(X_i \in L_b(x))}{|L_b(x)|}$$
 - C) Re-estimate ψ using weights on entire sample.
- → Difference from Causal Forest - trees are used for constructing weights!

Takeaways from today

- Delegation
 - AI first attempt and then subsequent only human is optimal
- Garden-based causal inference
 - Causal trees and forests
 - Unconfound using outcome similarity
 - Alternative to covariate-based matching
 - Heterogeneity function -
 - Generalized random forest
 - Unconfound and
- Next times
 - Double machine learning